Spark Plug Cleaner



Category: **Automotive** Type: **Full SOP** Priority: **High** 3 labs

Covers these lab items: Spark plug cleaning machine

What it is used for

A spark plug cleaning machine removes carbon deposits and other suitable contaminants from spark plugs using the cleaning method specified for the equipment. Correct plug inspection, controlled cleaning, and post-cleaning checks help support engine maintenance while reducing dust, eye, and equipment hazards.

Main parts

- Machine housing encloses the cleaning mechanism.
- Spark plug holder secures the plug during cleaning.
- Cleaning chamber contains the cleaning process where fitted.
- Compressed-air inlet supplies air to pneumatic models.
- Abrasive or cleaning medium removes deposits where the machine uses it.
- Control switch or valve starts and stops the cleaning cycle.
- Pressure regulator controls air pressure where fitted.
- Inspection port or work light helps examine the plug where available.

Before you start

1. Only trained personnel should operate the spark plug cleaning machine.
2. Read the machine instructions and confirm the cleaner is suitable for the spark plug type.
3. Inspect the housing, plug holder, lid, hose, fittings, controls, and power cable for damage.
4. Check that the cleaning medium and operating pressure match the manufacturer's requirements.
5. Inspect the spark plug for cracked ceramic, damaged threads, worn electrodes, or other defects that require replacement rather than cleaning.
6. Ensure dust extraction or ventilation is operating where required and wear suitable eye and respiratory protection.
7. Disconnect power or air before opening the chamber or servicing the machine.

How to use

1. Switch off the engine and allow the spark plug to cool before removing it from the engine.
2. Inspect and identify the spark plug and check whether it is suitable for cleaning.
3. Secure the plug in the holder according to the machine instructions.
4. Close the cleaning chamber or lid and confirm the safety interlock is engaged where fitted.
5. Set the approved cleaning duration and pressure or operating mode.

6. Start the cleaning cycle from the designated control and keep the chamber closed during operation.
7. Stop the machine if abnormal noise, leakage, excessive vibration, or a chamber fault occurs.
8. Wait for the cycle to finish and release pressure according to the instructions before opening the chamber.
9. Remove the plug carefully and remove residual cleaning medium using the approved method.
10. Inspect the ceramic insulator, threads, electrodes, and electrode gap against the engine manufacturer's specifications.
11. Replace the spark plug if damaged or outside the service limits; do not return an unserviceable plug to use.
12. Clean the machine, check for residue, and record any equipment or plug defects.

⚠ Safety

- Never operate the machine with the chamber open or required guards disabled.
- Wear suitable eye protection and respiratory protection where dust or cleaning media may become airborne.
- Do not clean a spark plug with cracked ceramic, damaged threads, or other defects that make it unserviceable.
- Disconnect power or air before clearing blockages or opening the chamber for maintenance.
- Keep hands away from moving parts and high-pressure air outlets.
- Follow the manufacturer's restrictions on cleaning media and do not use flammable or unapproved substances.

Care & storage

- Remove residual cleaning medium from the chamber after isolating the machine.
- Inspect the plug holder, lid, seals, hoses, fittings, and controls regularly.
- Replace cleaning media and service filters according to the manufacturer's schedule.
- Keep the machine clean and dry and store accessories in their designated locations.
- Arrange authorised service if pressure control, interlocks, or cleaning performance becomes unreliable.

Common mistakes to avoid

- Attempting to clean a spark plug that should be replaced.
- Opening the cleaning chamber before the cycle and pressure release are complete.
- Using the wrong cleaning medium or excessive operating pressure.
- Returning a plug to service without checking its ceramic, electrodes, and gap.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.